

The Generation of Spacetime: A Derivation Based on Angular-Momentum Network Geometry

时空的产生：一个从角动量网络几何学出发的推导

— Unified Origin of Time, Space, Force and Local Speed of Light
—— 时间、空间、力与局域光速的统一起源

Author: Chengbin Song

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Abstract

What is the “spacetime” perceived by human beings, and where does it originate? Within the framework of Angular-Momentum Network Geometry (ANG-TOE), this paper puts forward a complete derivation:

Spacetime is not a fundamental entity of the universe. It is a four-dimensional readout structure generated when angular-momentum quanta (S_g) project onto the human perceptual layer.

We demonstrate three core statements:

1. Fundamental cosmic layer (L0): space-free, time-free, speed-of-light-free, force-free. Only angular-momentum quanta and one governing rule exist: $\mathbf{J}_{\text{total}} \equiv 0$.

2. Human perceptual layer (L2): angular-momentum quanta are projected into four readouts: force, space, time and local speed of light.

3. Core locking equation:

This formula unifies the four perceptual quantities under one geometric constraint.

This paper deduces the generation mechanism of time, space, force and the local speed of light, and interprets the geometric nature of gravitational time dilation, dark matter and dark energy.

Keywords: origin of spacetime, angular-momentum quantum, local speed of light, topological density, ANG-TOE, time-exchange rate

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1. Introduction: Spacetime is not a “background”, but a “readout”

Since Newton’s era, physics has regarded spacetime as the “container” or background for physical events. Einstein upgraded this picture to “curved spacetime”, yet spacetime was still preserved as a fundamental physical quantity.

This paper proposes a more fundamental viewpoint: spacetime is not a primitive entity of the universe. It is a readout structure produced when angular-momentum quanta are projected onto the human perceptual layer.

At the fundamental layer L0:

- No space — no “position”, “distance” or “direction”.
- No time — no “sequence”, “flow rate” or “instant”.
- No speed of light — no “propagation velocity”.
- No force — no “push”, “pull” or “attraction”.

Only the following components exist:

- Angular-momentum quantum ($S_g = h$)
- Link network
- One rule: $J_{\text{total}} \equiv 0$

Spacetime emerges when this underlying structure is projected onto the human perceptual layer L2.

2. Fundamental Layer (L0): No spacetime, only angular momentum

2.1 The sole entity: link network

Within ANG-TOE, the only entity of the universe is the link network. Attributes of a single link are defined as:

Link = (J, ℓ, Θ, Φ)

J denotes angular-momentum flux, the only ontological quantity.

2.2 The unique rule: Axiom 0

- Global angular momentum must sum to zero.
- Any local angular momentum has to be counterbalanced by global counter-angular-momentum.
- This rule is topologically transient and does not presuppose time.

2.3 Quantization of angular momentum

S_g (Song unit) is the fundamental quantum of angular momentum, equivalent to Planck’s constant h .

3. Human Perceptual Layer (L2): The generation mechanism of spacetime

When angular-momentum quanta project from L0 to L2, they decompose into four human-accessible readouts:

3.1 Time = exchange rate of angular momentum

More precisely, under Axiom XI:

$$\Delta t = \frac{\bar{\ell}}{\bar{\kappa} \cdot \omega_{\text{topo}}}$$

ω_{topo} stands for the eigenfrequency of angular-momentum reconnection. A time step corresponds to one complete angular-momentum-exchange cycle.

Time is not an independently flowing quantity; it is the rhythm of angular-momentum exchange.

3.2 Space = positional unfolding of angular momentum

Equivalently, within the perceptual layer:

$$\mathbf{r} = \int \frac{\mathbf{J}}{|\mathbf{J}|} \cdot d\ell$$

Space is not a container. It is the readout generated by angular momentum unfolding along three translational directions.

3.3 Force = spatial gradient of angular momentum

- Force is neither push nor pull; it is the spatial change rate of angular momentum.
- Force equals zero when angular momentum distributes uniformly across space.
- Force takes a large value where angular momentum varies sharply in space.

3.4 Local speed of light = spacetime coupling coefficient

Where:

- c_0 is the benchmark vacuum speed of light (constant).
- $\Phi = |\text{Link}|/\chi$ represents local topological density.

The speed of light is not a universal constant. It is a modulation coefficient for the spacetime-coupling ratio determined by local topological density.

4. Unified locking equation: geometric constraint for spacetime product

Within layer L2, four perceptual quantities are locked into one single equation:

4.1 First equality: $r \cdot t = Sg/F$

- Derived from the classical definition $J = F \cdot r \cdot t$
- Substitute quantized angular momentum $J = Sg$ ($n = 1$)
- Interpretation: the spacetime product equals the ratio between the angular-momentum quantum and force.

4.2 Second equality: $r \cdot t = c(\Phi) \cdot t^2$

- Derived from $r = c(\Phi) \cdot t$ (local light-speed coupling)
- Interpretation: the spacetime product is modulated by local topological density Φ .

4.3 Complete closed chain

Parameter	Direction of variation	Influence on spacetime product
Angular-momentum quantum Sg	Fixed	Rigid anchor point
Force F	Increases	$r \cdot t$ decreases
Local topological density Φ	Increases	$c(\Phi)$ declines, altering $r \cdot t$
Time t	Varies	Space r varies synchronously to keep the product stable

5. Application: geometric interpretation of gravitational time dilation

In general relativity, time slows down within a strong gravitational field. Within the ANG framework, the mechanism is listed below:

Step	Mechanism	Result
1	Approaching massive objects $\rightarrow$ local topological density Φ rises	—
2	Growth of $\Phi \rightarrow$ reduction of local speed of light $c(\Phi)$	Theorem T10
3	$r \cdot t = c(\Phi) \cdot t^2$, the spacetime product gets modulated	—
4	Given fixed spatial scale r	Time t has to adjust

5	Human perception of “time dilation”	Reduced time flow rate in strong gravitational fields
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Gravitational time dilation does not arise from spacetime curvature. It is the redistribution of the spacetime product induced by scaled local light speed.

6. Application: geometric nature of dark matter and dark energy

6.1 Dark matter

Dark matter represents residual-force readouts on the spatial projection of angular-momentum halos ($\text{Link}_{\text{halo}} \propto 1/r$). It does not exist as a primitive substance on layer L0. It is incompletely-read angular-momentum information on perceptual layer L2.

6.2 Dark energy

Dark energy corresponds to expansion-readouts on the temporal projection generated by global angular-momentum-cancellation residuals. It does not “push” the universe outward. It is the L2-layer misreading of boundary drift on layer L0.

7. Falsifiable predictions

Prediction	Falsification condition	Test method
Local speed of light $c(\Phi)$ varies with topological density Φ	c proves independent of Φ	High-precision atomic-clock comparison
Relation $r \cdot t = Sg/F$ holds across all scales	Deviation larger than 1%	Precise mechanical-temporal coupling experiments
Dark-matter density is proportional to $1/r^2$	Observed deviation beyond 3σ	Galactic rotation-curve measurement
Gravitational time dilation originates from light-speed scaling rather than spacetime curvature	Non-GR spacetime readout patterns detected	Gravitational-wave observation

8. Conclusion

Only angular momentum exists at the fundamental level of the universe. Spacetime is a shadow-structure acquired by humans within the projection layer. Force, space, time and the speed of light are all unfolded readouts of angular-momentum quanta along four distinct dimensions. ©  ✨

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End of Paper

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作者: Chengbin Song

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摘要

人类感知的“时空”是什么？它从何而来？本文在角动量网络几何学（ANG-TOE）框架中，提出一个完整的推导：

时空不是宇宙的底层实体，而是角动量量子（Sg）投影到人类感知层时产生的四维读数结构。

我们证明：

- 宇宙底层（L0）：无空间，无时间，无光速，无力。只有角动量量子 and 一条规则： $\mathbf{J}_{\text{total}} \equiv 0$ 。
- 人类感知层（L2）：角动量量子被投影为力、空间、时间、局域光速四个读数。
- 核心锁等式：

该式将所有四个感知量统一在同一个几何约束下。

本文推导了时间、空间、力、局域光速的产生机制，并解释了引力时间膨胀、暗物质暗能量的几何本质。

关键词：时空起源、角动量量子、局域光速、拓扑密度、ANG-TOE、时间交换率

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1. 引言：时空不是“背景”，它是“读数”

物理学自牛顿以来，一直将“时空”视为物理事件的“容器”或“背景”。爱因斯坦将其升级为“时空弯曲”，但依然保留了时空作为物理量的核心地位。

本文提出一个更根本的观点：时空不是宇宙的底层实体，它是角动量量子投影到人类感知层时产生的读数结构。

在宇宙底层 (L0)：

- 没有空间 —— 没有“位置”、“距离”、“方向”。
- 没有时间 —— 没有“先后”、“流速”、“瞬间”。
- 没有光速 —— 没有“传播速度”。
- 没有力 —— 没有“推”、“拉”、“吸引”。

只有：

- 角动量量子 ($S_g = h$)
- 链接网络
- 一条规则： $\mathbf{J}_{\text{total}} \equiv 0$

当这个底层结构投影到人类感知层 (L2) 时，时空被“产生”了出来。

2. 宇宙底层 (L0)：无时空，只有角动量

2.1 唯一实体：链接网络

在 ANG-TOE 中，宇宙的唯一实体是链接网络。每个链接的属性：

链接 = $(\mathbf{J}, \ell, \Theta, \Phi)$

$\mathbf{J}$ 是角动量通量，是唯一的“本体量”。

2.2 唯一规则：Axiom 0

- 全局角动量必须归零。
- 任何局域角动量必须被全局反角动量抵消。
- 这个规则是拓扑瞬态的 —— 不涉及时间。

2.3 角动量量子化

Sg（宋单位）是角动量的基本量子，等于普朗克常数 h 。

3. 人类感知层（L2）：时空的产生机制

当角动量量子从 L0 投影到 L2 时，它被“拆解”为四个人类可感知的读数：

3.1 时间 = 角动量交换速率

更精确地，根据 Axiom XI：

$$\Delta t = \frac{\bar{\ell}}{\bar{\kappa} \cdot \omega_{\text{topo}}}$$

ω_{topo} 是角动量重联的本征频率。时间步长是角动量交换一次所需的时间。

时间不是“流逝”的独立量，它是角动量交换的“节奏”。

3.2 空间 = 角动量的位置展开

或等价地，在感知层中：

$$\mathbf{r} = \int \frac{\mathbf{J}}{|\mathbf{J}|} \cdot d\ell$$

空间不是容器，它是角动量在三个平移方向上的展开读数。

3.3 力 = 角动量的空间梯度

- 力不是推或拉，它是角动量在空间方向上的变化率。
- 当角动量在空间中均匀分布时，力为零。
- 当角动量在空间中急剧变化时，力很大。

3.4 局域光速 = 时空耦合系数

其中：

- c_0 是真空基准光速（常数）。
- $\Phi = |\text{Link}|/\chi$ 是局域拓扑密度。

光速不是宇宙的常数，它是局域拓扑密度对时空耦合比例的调制系数。

4. 统一锁等式：时空乘积的几何约束

在 L2 层，四个感知量被锁定在同一个等式中：

4.1 第一等式： $r \cdot t = Sg/F$

- 来自经典定义 $J = F \cdot r \cdot t$
- 代入量子化 $J = Sg$ （取 $n = 1$ ）
- 含义：时空乘积是角动量量子与力的比值。

4.2 第二等式： $r \cdot t = c(\Phi) \cdot t^2$

- 来自 $r = c(\Phi) \cdot t$ （局域光速耦合）
- 含义：时空乘积受局域拓扑密度 Φ 调制。

4.3 完整闭合链

参数	变化方向	对时空乘积的影响
角动量量子 Sg	固定	刚性锚点
力 F	增大	$r \cdot t$ 减小
局域拓扑密度 Φ	增大	$c(\Phi)$ 减小, $r \cdot t$ 改变
时间 t	变化	空间 r 同步变化（保持乘积恒定）

5. 应用：引力时间膨胀的几何解释

在广义相对论中，强引力场使时间变慢。在 ANG 中：

步骤	机制	结果
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1	靠近大质量天体 → 局域拓扑密度 Φ 增大	—
2	Φ 增大 → 局域光速 $c(\Phi)$ 减小	T10 定理
3	$r \cdot t = c(\Phi) \cdot t^2$ 时空乘积被调制	—
4	若空间 r 被“钉住”	时间 t 必须变化
5	人类感知到“时间膨胀”	强引力场中时间流速降低

引力时间膨胀不是时空弯曲，而是局域光速缩放导致时空乘积重分布。

6. 应用：暗物质暗能量的几何本质

6.1 暗物质

暗物质是角动量尾迹 ($\text{Link}_{\text{halo}} \propto 1/r$) 在空间投影上的残余力读数。它不“存在”于 L0，它是人类在 L2 中漏读的角动量信息。

6.2 暗能量

暗能量是全局角动量对消残余在时间投影上的膨胀读数。它不“推动”宇宙，它是 L2 层对 L0 边界漂移的误读。

7. 可证伪预测

预测	证伪条件	检验方式
局域光速 $c(\Phi)$ 随拓扑密度 Φ 变化	c 与 Φ 无关	高精度原子钟比对
$r \cdot t = Sg/F$ 在所有尺度成立	偏差 $> 1\%$	精密力学-时间耦合实验
暗物质密度与 $1/r^2$ 成比例	观测偏离 $> 3\sigma$	星系旋转曲线
引力时间膨胀不是时空弯曲，而是光速缩放	探测到非 GR 的时空读数模式	引力波探测

8. 结论

宇宙底层只有角动量。时空是人类在投影层中读到的“影子结构”。力、空间、时间、光速 —— 都是角动量量子在四个不同方向上的展开读数。🌀📐⚡

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